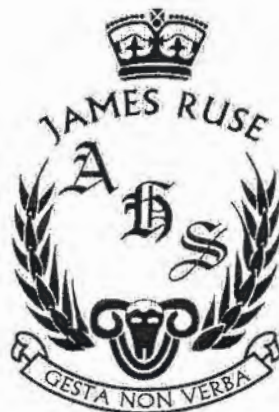


Name:	
Class:	



YEAR 12

ASSESSMENT TEST 2

TERM 1, 2015

MATHEMATICS

EXTENSION 2

Time Allowed – 120 Minutes
(Plus 5 minutes Reading Time)

General Instructions:

- All questions may be attempted
- All questions are of equal value
- Standard Integral Tables will be supplied
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper

Question 1-5 are to be completed on Multiple choice sheet then Question 6,7,8,9 to be completed on separate sheets of paper and handed in in separate bundles. Each question must show your (in the top right hand corner) Candidate Number.

YEAR 12 – 2015 – EXTENSION 2 TERM 1 -TASK 2
Use the multiple choice answer sheet supplied for question 1 to 5

- Q1 The area of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ in units square is
 A. 36π B $36\pi^2$ C 6π D $6\pi^2$
-
- Q2 Given any point $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.
 The order of the quadrants on a circle in which $P(a \sec \theta, b \tan \theta)$ from $\theta = 0$ to 2π runs is:
 A. 1, 2, 3, 4 B. 1, 4, 2, 3 C. 1, 3, 2, 4 D. 1, 4, 3, 2
-
- Q3 The equation of the tangent at a point $P(x_0, y_0)$ on the curve: $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ in the first quadrant is
 A. $y_0^{\frac{1}{3}}x + x_0^{\frac{1}{3}}y = x_0^{\frac{1}{3}}y_0^{\frac{1}{3}}a^{\frac{1}{3}}$.
 B. $y_0^{\frac{1}{3}}x + x_0^{\frac{1}{3}}y = x_0^{\frac{1}{3}}y_0^{\frac{1}{3}}a^{\frac{4}{3}}$.
 C $y_0^{\frac{1}{3}}x + x_0^{\frac{1}{3}}y = 2x_0^{\frac{1}{3}}y_0^{\frac{1}{3}}$.
 D. $y_0^{\frac{1}{3}}x + x_0^{\frac{1}{3}}y = x_0^{\frac{1}{3}}y_0^{\frac{1}{3}}a^{\frac{2}{3}}$.
-
- Q4 Using an appropriate substitution
 $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec^2 x}{(1 + \tan x)^2} dx$ is equivalent to
 A. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u} du$ C. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u^2} du$
 B. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{u^2}{(1+u)^3} du$ D. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u^3} du$
-
- Q5. The equation $|z - 1 - 3i| + |z - 9 - 3i| = 10$ corresponds to an ellipse in the Argand diagram. Which of the following is the complex number corresponding to the centre of the ellipse?
 A. $5 + 3i$ B. $-5 + 3i$
 C. $-5 - 3i$ D. $5 - 3i$

QUESTION 6 (20 Marks)

a) $\int_{\frac{\pi}{16}}^{\frac{\pi}{12}} \sec 4x \tan 4x \, dx$ 2

b) Find $\int \frac{1}{\sqrt{x^2 - 6x + 34}} dx$ 2

c) (i) Find real numbers a, b and c such that

$$\frac{3x}{(x+1)(x^2+2x+4)} \equiv \frac{a}{x+1} + \frac{bx+c}{x^2+2x+4}$$
 2

(ii) Find $\int \frac{3x}{(x+1)(x^2+2x+4)} dx$ 2

d) Let $t = \tan \frac{\theta}{2}$

(i) Show that $d\theta = \frac{2}{1+t^2} dt$ 1

(ii) Use the substitution $t = \tan \frac{\theta}{2}$ to find

$$\int \frac{1}{3 \sin \theta + 4 \cos \theta + 5} d\theta$$

e) On separate number planes draw sketches of the following, for $-2\pi \leq x \leq 2\pi$.

(i) $y^2 = \sin x$ 2

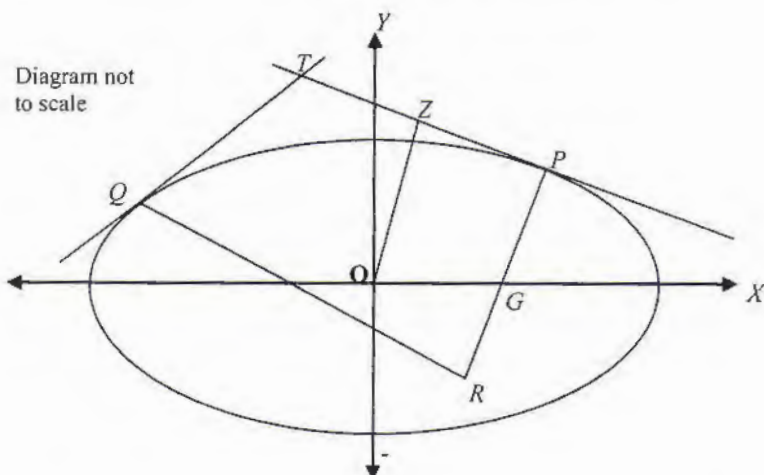
(ii) $|y| = \sin x$ 2

(iii) $y = (\sin^3 x)$ 2

f) If z is any point on the circle $|z-1| = 1$ prove that $\arg(z-1) = 2\arg z$ 2

QUESTION 7 START A NEW PAGE (20 Marks)

- a) Consider the ellipse $E: x^2 + 4y^2 = 100$. PT and QT are tangents at $P(6, 4)$ and $Q(-8, 3)$ respectively. They meet at the point T .



- (i) Show that the equation of PT is given by $3x + 8y = 50$. 2
- (ii) PR and QR are normals at P and Q respectively.
Show the equation of PR is given by $8x - 3y = 36$. 2

PR meets the major axis in G , and OZ is the perpendicular from the centre O to the tangent at P .

- (iii) Prove that $PG \cdot OZ = 25$. 3
- (iv) Find the coordinates of T and R . 4
- (v) Show that the diameter through R is perpendicular to PQ . 2

b)

- (i) Solve $z^3 = \sqrt{2} + \sqrt{2}i$, giving answers in the form $R \operatorname{cis} \theta$. 2
- (ii) Hence prove that $\cos\left(\frac{\pi}{12}\right) + \cos\left(\frac{9\pi}{12}\right) + \cos\left(\frac{-7\pi}{12}\right) = 0$ 1

- (c) Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 x \sin^5 x \, dx$ 1

- (d) Find $\int \sqrt{x} \ln x \, dx$ 3

QUESTION 8 START A NEW PAGE (20 Marks)

a) Given the complex number $z = \cos \theta + i \sin \theta$

(i) Show that $Z^n + \frac{1}{Z^n} = 2 \cos n\theta$ 2

(ii) Hence by expanding $(Z + \frac{1}{Z})^4$, find an expression for $\cos^4 \theta$ in the form
 $a \cos 4\theta + b \cos 2\theta + c$ 3

(iii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta$ 2

b) (i) Sketch $y = \frac{x^2}{x^3 + 1}$, showing all the essential points 4

(ii) Hence find the number of real roots for $x^3 - 5x^2 + 1 = 0$. 2

c) Let $I_n = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2nx}{\sin 2x} dx$ for $n=1, 2, 3, \dots$

(i) Evaluate I_1 2

Using the fact that $\cos S - \cos T = 2 \sin\left(\frac{S+T}{2}\right) \sin\left(\frac{S-T}{2}\right)$.

(ii) Show that for $r \geq 1$, $I_{2r+1} - I_{2r-1} = \frac{1 - (-1)^r}{2r}$ 3

(iii) Hence evaluate I_7 2

QUESTION 9 **START A NEW PAGE (20 MARKS)**

a)

(i) Show that $\int_{-a}^a f(x)dx = \int_{-a}^a f(-x)dx$ 1

(ii) Hence or otherwise evaluate $\int_{-4}^4 (e^x - e^{-x}) \cos x dx$ 2

(iii) By considering $\int_0^a \sqrt{a^2 - x^2} dx$ find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 2

b)

(i) Sketch the hyperbola $H: \frac{x^2}{9} - \frac{y^2}{16} = 1$ showing all the asymptotes, directrices, vertices and foci. 3

(ii) Show the equation of the tangent at any point $P(3 \sec \theta, 4 \tan \theta)$ on the hyperbola H is $4x \sec \theta - 3y \tan \theta = 12$. 2

The tangent at P meets the asymptotes at Q in the first quadrant and R in the fourth quadrant.

(iii) If O is the centre of H , prove that P is the mid-point of QR . 4

(iv) Find the area of $\triangle OQR$. 3

(v) If S is the focus of H , find $\angle QSR$ to the nearest minute. 3

MATHEMATICS Extension 2: Question 1-5

Suggested Solutions	Marks	Marker's Comments
Q1 Area is πab $\therefore \pi \times 3 \times 2 = 6\pi$ part C	1	
Q2 $\sec \theta$ positive in 1st and 4th $\tan \theta$ positive in 1st and 3rd. $(+, +), (-, -), (-, +), (+, -)$ 1st, 3rd, 2nd, 4th. part C.	1	
Q3 $y_0^{1/3} x + x_0^{1/3} y = \frac{2}{3} x_0^{1/3} y_0^{1/3} a^{2/3}$ part D see separate sheet.	1	
Q4 let $u = 1 + \tan x$ $\frac{dy}{dx} = \sec^2 x$ $x = \pi/4 \quad u = 2$ $x = -\pi/4 \quad u = 0$ $\int_0^2 \frac{1}{u^2} du$ most correct answer part C	1	
From $(1, 3)$ to $(9, 3)$ midpoint is $(5, 3)$ so centre is <u>$5 + 3i$</u>		

$$a) \left[\frac{1}{4} \sec 4x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \frac{1}{4} \left(\sec \frac{\pi}{3} - \sec \frac{\pi}{4} \right)$$

$$= \frac{1}{4} (2 - \sqrt{2}) \#$$

1 some forget $\frac{1}{4}$

$$b) \int \frac{dx}{\sqrt{(x-3)^2 + 25}}$$

$$= \ln(x-3 + \sqrt{(x-3)^2 + 25}) + c \#$$

1

1

1

$$c) \frac{3x}{(x+1)(x^2+2x+4)} = \frac{a}{x+1} + \frac{bx+c}{x^2+2x+4}$$

$$3x = a(x^2+2x+4) + (bx+c)(x+1)$$

$$x=0, 0=4a+c \quad (1)$$

$$x=-1, -3=3a \quad \therefore a=-1$$

$$\text{Sub in } (1) \quad c=4$$

$$x=1, 3=7a+2b+2c$$

$$3=-7+2b+8$$

$$1=b$$

$$a=-1, b=1, c=4$$

Any 2 correct 1m

All correct 2m

$$i) \int \left(\frac{-1}{x+1} + \frac{x+4}{x^2+2x+4} \right) dx$$

$$= -\ln(x+1) + \frac{1}{2} \int \frac{2x+2}{x^2+2x+4} dx + \int \frac{3dx}{x^2+2x+4}$$

$$= -\ln(x+1) + \frac{1}{2} \ln(x^2+2x+4) + \sqrt{3} \tan^{-1} \left(\frac{x+1}{\sqrt{3}} \right) + c \quad 1m$$

1m

a lot of students got $\sqrt{3}$ wrong.

#.

$$d) i) \frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$$

$$= \frac{1}{2} (1 + \tan^2 \frac{\theta}{2})$$

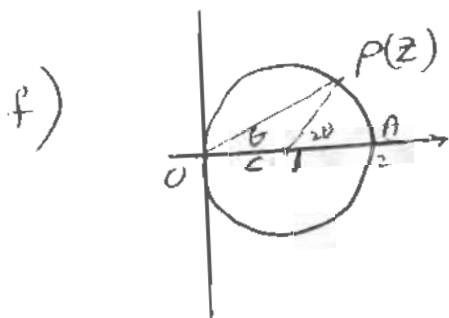
$$= \frac{1}{2} (1 + t^2)$$

$$\frac{d\theta}{dt} = \frac{2}{1+t^2}$$

$$ii) \int \frac{1}{3 + \frac{2t}{1+t^2} + \frac{4(1-t^2)}{1+t^2} + 5} \cdot \frac{2}{1+t^2} dt$$

$$= \int \frac{2}{(t+3)^2} dt$$

$$= -\frac{2}{\tan \frac{\theta}{2} + 3} dt$$



$$(x-1)^2 + y^2 = 1$$

centre (1, 0) at C

$$\text{Let } \angle POA = \theta = \arg(z)$$

$$\angle PCA = \arg(z-1)$$

$\angle PCA = 2\theta$ (angle at centre
twice angle at
circumference standing
on same arc)

$$\therefore \arg(z-1) = 2\arg(z)$$

#

1 m

1 m for substitution

1 m complete square

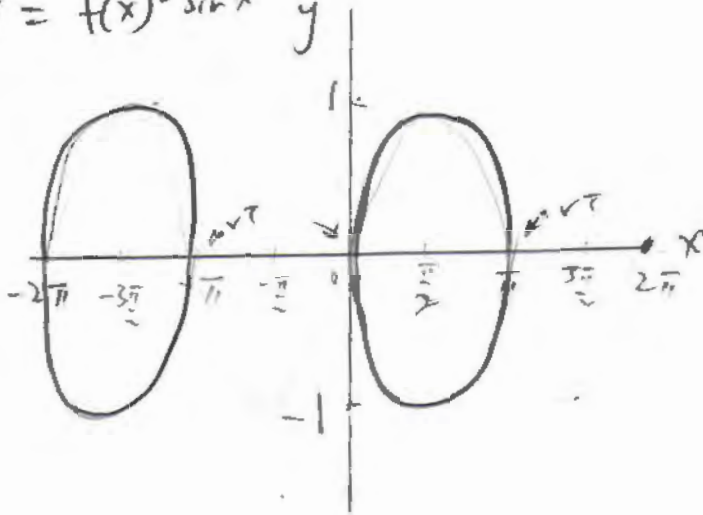
1 m
some forget to change
back to θ

1 m
must introduce
where is $\arg(z)$
 $\arg(z+1)$

1 m
some forget on same
arc did not get full
mark

6e) $-2\pi \leq x \leq 2\pi$

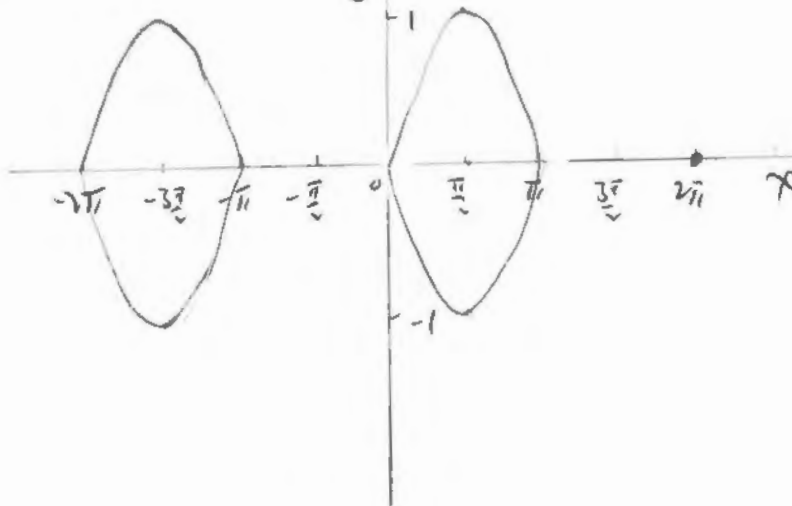
i) $y^2 = f(x) = \sin x$



1m for 2 ovals
in the right place

1m for vertical
tangents
must locate TP.

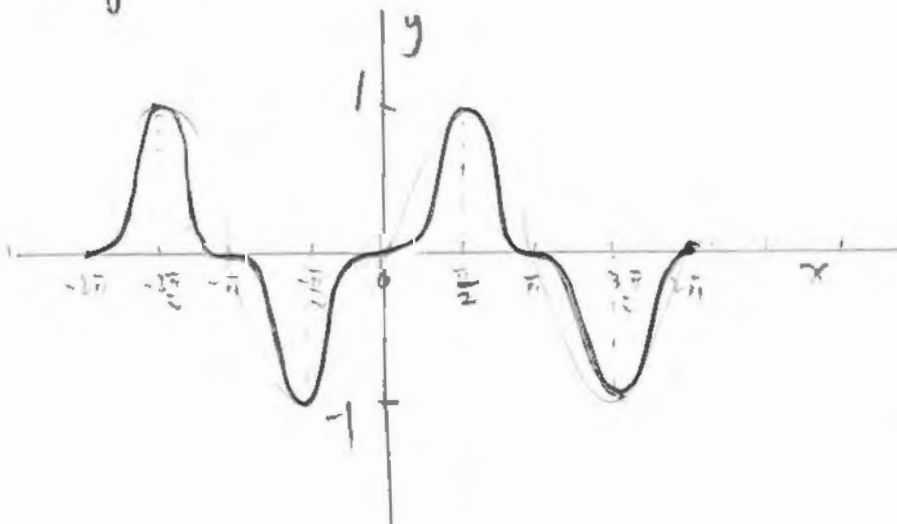
ii) $|y| = f(x) = \sin x$



1m for 2 ovals
in the right place
1m for correct shape
must locate TP

(i) & (ii) must be
different in shape

iii) $y = f^3(x) = \sin^3 x$



badly done

1m for correct
shape with TP
(Concave in)

1m for HPOI
cutting the x-axis

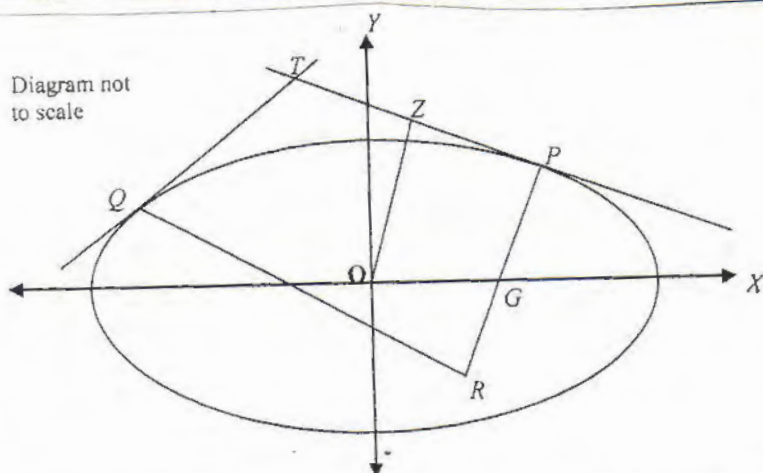
MATHEMATICS Extension 2: Question...7....

Suggested Solutions

Marks

Marker's Comments

(a)



(i) $x^2 + 4y^2 = 100$

$2x + 8y \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{x}{4y}$ at $x=6$ $y=4$ $m_T = -3/8$

Equation of tangent at P

$y - 4 = -\frac{3}{8}(x - 6)$

$8y - 32 = -3x + 18$

$3x + 8y = 50$

(ii) $m_N = \frac{8}{3}$ Equation of normal at P

$y - 4 = \frac{8}{3}(x - 6)$

$3y - 12 = 8x - 48$

$\therefore 8x - 3y = 36$

(iii) when $y = 0$ and $8x - 3y = 36$

$\therefore x = 9/2$

$\therefore G = (9/2, 0)$

$PG = \frac{|3(9/2) + 8(0) - 50|}{\sqrt{3^2 + 8^2}} = \frac{73}{2\sqrt{73}}$

$OZ = \frac{|3(0) + 8(0) - 50|}{\sqrt{3^2 + 8^2}} = \frac{50}{\sqrt{73}}$

$\therefore PG \times OZ = \frac{73}{2\sqrt{73}} \times \frac{50}{\sqrt{73}} = 25$

②

① gradient of tangent

① straight line equation

②

① gradient straight line equation

* can not just quote equations of tangent or normal to ellipse

③

① $G(9/2, 0)$

① PG

① OZ and answer.

MATHEMATICS Extension 2: Question 7

Suggested Solutions	Marks	Marker's Comments
b) (i) $z^3 = \sqrt{2} + \sqrt{2}i = 2 \text{cis}(\pi/4)$ let $z = R \text{cis} \theta$ $R^3 \text{cis} 3\theta = 2 \text{cis} \pi/4$ (By De Moivre) $R = \sqrt[3]{2}$ $3\theta = \pi/4 + 2k\pi$ $k \in \mathbb{Z}$ $\theta = \frac{\pi + 6k\pi}{12}$ $k=0 \therefore z_1 = \sqrt[3]{2} \text{cis}(\pi/12)$ $k=1 \quad z_2 = \sqrt[3]{2} \text{cis}(\frac{9\pi}{12}) = \sqrt[3]{2} \text{cis}(\frac{3\pi}{4})$ $k=-1 \quad z_3 = \sqrt[3]{2} \text{cis}(\frac{-7\pi}{12})$ $z^3 - 2 \text{cis}(\pi/4) = 0$ sum of roots = 0 coefficient of z^2 term $\sqrt[3]{2} \text{cis}(\pi/12) + \sqrt[3]{2} \text{cis}(\frac{3\pi}{4}) + \sqrt[3]{2} \text{cis}(\frac{-7\pi}{12}) = 0$ Equate Real parts $\therefore \cos(\pi/12) + \cos(\frac{3\pi}{4}) + \cos(\frac{-7\pi}{12}) = 0$ (c) $\int_{-\pi/2}^{\pi/2} \cos^4 x \sin^5 x \, dx = 0$ $-\pi/2$ as $\cos^4 x \sin^5 x$ is an odd function (d) $\int \sqrt{x} \ln x \, dx$ $= \left[\frac{2}{3} x^{3/2} \ln x \right] - \frac{2}{3} \int x^{3/2} \cdot \frac{1}{x} \, dx$ $= \frac{2}{3} x \sqrt{x} \ln x - \frac{2}{3} \int x^{1/2} \, dx$ $= \frac{2}{3} x \sqrt{x} \ln x - \frac{2}{3} \times \frac{2}{3} x^{3/2} + C$ $= \frac{2}{3} x \sqrt{x} \ln x - \frac{4}{9} x \sqrt{x} + C$	(2) (1) (1) (1) (1) (3)	(1) $z_1 = \sqrt[3]{2} \text{cis} \pi/4$ (1) All 3 solutions (1) state sum of roots = 0 write out sum of roots state equate real part. (1) Must give reason. (no need to show any integration) (2) integration by parts. (1) Answer.

MATHEMATICS Extension 2: Question...7...

Suggested Solutions	Marks	Marker's Comments
(iv) Equation of QT tangent at Q $m = \frac{-(-8)}{4(3)} = \frac{8}{12} = \frac{2}{3}$ $y - 3 = \frac{2}{3}(x + 8)$ $3y - 9 = 2x + 16$ To find T $\begin{aligned} 2x - 3y &= -25 & (i) \\ 3x + 8y &= 50 & (ii) \\ 6x - 9y &= -75 \\ 6x + 16y &= 100 \\ -25y &= -175 \\ y &= 7 \\ x &= -2 \end{aligned}$ T = (-2, 7) Equation of Normal at Q $m = -\frac{3}{2}$ $y - 3 = -\frac{3}{2}(x + 8)$ $2y - 6 = -3x - 24$ $\begin{aligned} 3x + 2y &= -18 & (i) \\ 8x - 3y &= 36 & (ii) \\ 9x + 6y &= -54 \\ 16x - 6y &= 72 \\ 25x &= 18 \\ x &= \frac{18}{25} \\ y &= \frac{-18 - 3(\frac{18}{25})}{2} \\ &= \frac{-252}{25} \end{aligned}$ R = $\left[\frac{18}{25}, \frac{-252}{25}\right]$	(4)	① Equation of QT ① T (-2, 7) ① Equation of QR. ① R $\left[\frac{18}{25}, \frac{-252}{25}\right]$
(v) Gradient of PQ = $\frac{4-3}{6-(-8)} = \frac{1}{4}$ Gradient of OR = $-\frac{252}{25} / \frac{18}{25} = -14$ $\therefore m_{PQ} \times m_{OR} = \frac{1}{4} \times -14$ $= -\frac{7}{2}$ $\therefore PQ$ is perpendicular to OR		① $m_{PQ} = \frac{1}{4}$ ① m_{OR} and proof of perpendicular

MATHEMATICS Extension 2: Question 8

Suggested Solutions	Marks	Marker's Comments
(a) (i) Given that $z = \cos \theta + i \sin \theta$ $\text{LHS} = z^n + \frac{1}{z^n}$ $= (\cos \theta + i \sin \theta)^n + \frac{1}{(\cos \theta + i \sin \theta)^n}$ $= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta) \quad \text{ De Moivre's Theorem}$ $= \cos n\theta + i \sin n\theta + \cos(n\theta) - i \sin n\theta \quad \text{ (cos is an even fn, sin is an odd fn)}$ $= 2 \cos n\theta$ $= \text{RHS}$		
(ii) $(z + \frac{1}{z})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$ (Binomial expansion) $(z + \frac{1}{z})^4 = (z^2 + \frac{1}{z^2}) + 4(z^2 + \frac{1}{z^2}) + 6$ $(2 \cos \theta)^4 = 2 \cos 4\theta + 4(2 \cos 2\theta) + 6$ $16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$ $\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$		
(iii) $\int_0^{\pi/2} \cos^4 \theta d\theta = \int_0^{\pi/2} (\frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}) d\theta$ $= \left[\frac{\sin 4\theta}{32} + \frac{\sin 2\theta}{4} + \frac{3\theta}{8} \right]_0^{\pi/2}$ $= \left(\frac{\sin 2\pi}{32} + \frac{\sin \pi}{4} + \frac{3\pi/2}{8} \right) - (0)$ $= \frac{3\pi}{16}$		
(b) (i) $y = \frac{x^2}{x^3+1}$ $\frac{dy}{dx} = \frac{2x(x^3+1) - x^2(3x^2)}{(x^3+1)^2} \quad x \neq -1$ $= \frac{2x^4 + 2x - 3x^4}{(x^3+1)^2}$ $= \frac{x(2-x^3)}{(x^3+1)^2}$		

MATHEMATICS Extension 2: Question 8...

Suggested Solutions

Marks

Marker's Comments

stat pt at $x = \sqrt[3]{2}$ and $x = 0$

$$\frac{dy}{dx} = \frac{(2-4x^3)(x^3+1)^2 - (2x-x^4)(2 \cdot 3x^2(x^3+1))}{(x^3+1)^4}$$

$$= \frac{(2-4x^3)(x^3+1) - 6x^2(2x-x^4)}{(x^3+1)^3}$$

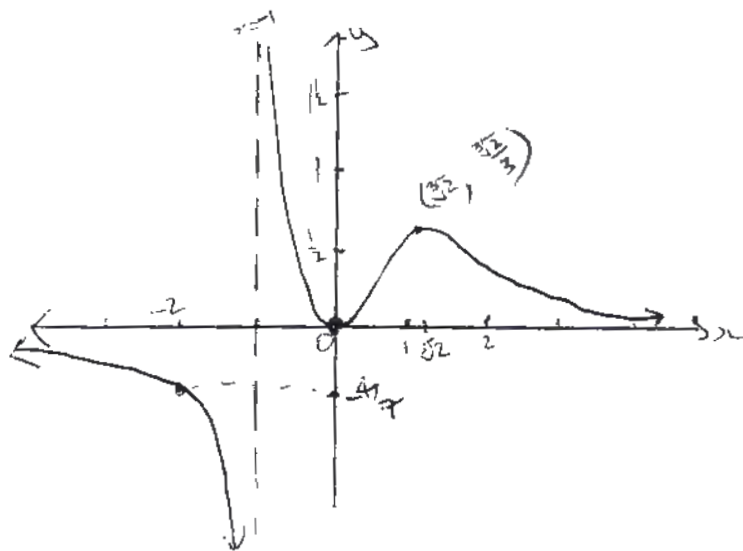
when $x = \sqrt[3]{2}$; $\frac{dy}{dx} = \frac{-5(3) - 6\sqrt[3]{2}^2(2\sqrt[3]{2} - \sqrt[3]{2} \cdot 2)}{27}$

$$= -5/9$$

concave down

∴ rel. max at $(\sqrt[3]{2}, 0.529)$

$(0,0)$ horizontal asymp. at $y=0$
vertical asymp. at $x=-1$



b)(iii) $x^3 - 5x^2 + 1 = 0$

intersection between $y = \frac{x^2}{x^3+1}$ and

$y = \frac{1}{5}$, from our graph we can

see there would be 3 solutions.

when $x=0$,

$$\frac{dy}{dx} = 2$$

∴ rel min at $(2, 4/9)$

1 for data

1x for each branch
drawn correctly

or If calculus wasn't
used at all, 1 for each
branch, 1 for vert. asymp
and 1 for stat. pts.

1 for stating the 2
graphs

1 for correct

MATHEMATICS Extension 2: Question 8

Suggested Solutions	Marks	Marker's Comments
(i) $I_1 = \int_0^{\pi/4} \frac{1 - \cos 2x}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{1 - \cos^2 x + \sin^2 x}{2 \sin x \cos x} dx$ $= \int_0^{\pi/4} \frac{2 \sin^2 x}{2 \sin x \cos x} dx$ $= \int_0^{\pi/4} \frac{\sin x}{\cos x} dx$ $= \left[-\ln \cos x \right]_0^{\pi/4}$ $= -\ln \frac{1}{\sqrt{2}} + \ln 1$ $= \ln \sqrt{2} = \frac{1}{2} \ln 2$	1	
(ii) $I_{2r+1} - I_{2r-1}$ $= \int_0^{\pi/4} \frac{1 - \cos 2(2r+1)x}{\sin 2x} dx - \int_0^{\pi/4} \frac{1 - \cos 2(2r-1)x}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{1 - \cos 2(2r+1)x - 1 + \cos 2(2r-1)x}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{\cos (4r-2)x - \cos (4r+2)x}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{2 \sin \left(\frac{4rx - 2x + 4rx + 2x}{2} \right) \sin \left(\frac{4rx - 2x - 4rx - 2x}{2} \right)}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{-2 \sin (4rx) \sin (-2x)}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{2 \sin (4rx) \sin 2x}{\sin 2x} dx \quad (\text{as } \sin x \text{ is odd fn})$ $= 2 \int_0^{\pi/4} \sin (4rx) dx$	1	if you didn't show this or explain why the minus sign goes then you lost a mark

MATHEMATICS Extension 2 Question 8...

Suggested Solutions	Marks	Marker's Comments
$= 2 \left[\frac{\cos 4rx}{4r} \right]_0^{6/4}$ $= \frac{1}{2r} [\cos r\pi - \cos 0]$ $= \frac{1}{2r} (-1^r - 1)$ $= \frac{1}{2r} (1 - (-1)^r)$ $= \frac{1 - (-1)^r}{2r}$	1	
(ii) so $I_{2r+1} - I_{2r-1} = \frac{1 - (-1)^r}{2r}$ when $r=3$; $I_7 - I_5 = \frac{1 - (-1)^3}{6} = \frac{1}{3}$ when $r=2$; $I_5 - I_3 = \frac{1 - 1}{4} = 0$ when $r=1$; $I_3 - I_1 = \frac{2}{2} = 1$		1
now $I_7 - I_5 + I_5 - I_3 + I_3 - I_1 = \frac{1}{3} + 0 + 1$ $\therefore I_7 - I_1 = \frac{4}{3}$ $I_7 = \frac{4}{3} + I_1$ $= \frac{4}{3} + \frac{1}{2} \ln 2$		1

MATHEMATICS Extension 2: Question 9

Suggested Solutions

Marks

Marker's Comments

2) Let $x = -u$

then $x = a \quad u = -a$

$x = -a \quad u = a$

$\int_a^{-a} f(-u) \frac{dx}{du} = -1$

$= \int_{-a}^a f(-u) du$

$= \int_{-a}^a f(-x) dx$ (use of a dummy variable)

Students make incorrect assumptions about being odd or even.

1

4) $I = \int_{-4}^4 (e^x - e^{-x}) \cos x dx$

using (1)

$= \int_{-4}^4 e^{-x} - e^x \cos(-x) dx$

$= - \int_{-4}^4 e^x - e^{-x} \cos x dx$

Since $\cos x$ is an even fn

$\therefore I = -I$

$2I = 0$

$I = 0$

1

1

A number of students obtained $\int_{-4}^4 0 dx$ then stated this was $\int_{-4}^4 0 dx$

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

a) $\int_0^a \sqrt{a^2 - x^2} dx$ is a quarter of a circle radius a

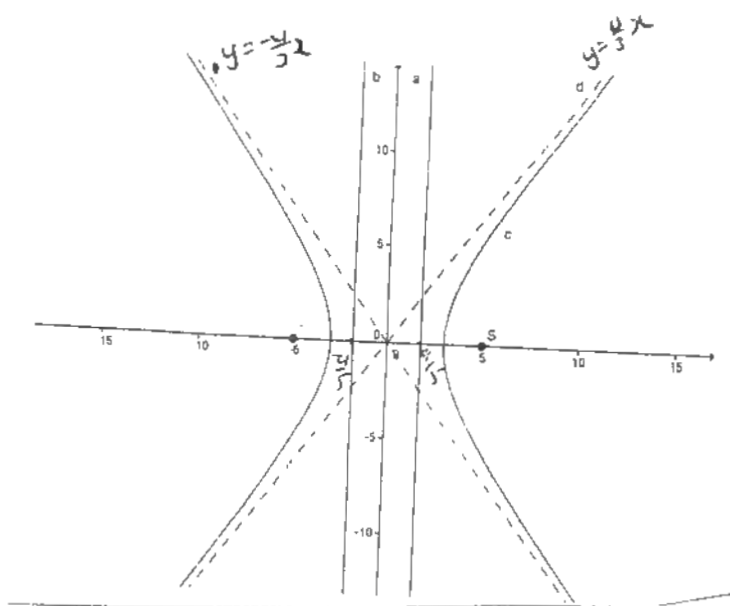
ii) $\int_0^a \sqrt{a^2 - x^2} dx = \frac{1}{4} \pi a^2$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \therefore y^2 = b^2 \left[1 - \frac{x^2}{a^2} \right]$

$y = \pm \frac{b}{a} [a^2 - x^2]$

$\therefore \text{area} = \frac{4b}{a} \int \sqrt{a^2 - x^2} dx$
 $= \frac{1}{4} \times \pi a^2 \times \frac{b}{a} \times 4$
 $= \pi ab$

(b)



$16 = 9(e^2 - 1)$

$\therefore \frac{16}{9} = e^2 - 1$
 $= \frac{25}{9}$

$\therefore x = \frac{9}{5} \text{ or } -\frac{9}{5}$

$y = \pm \frac{4}{3} x$

$S(5, 0) \quad S'(-5, 0)$

Poorly done for a standard question.

Many students used trig. substitution, which wasn't needed

1 for e
 1 for asymptotes and directions
 1 for graph

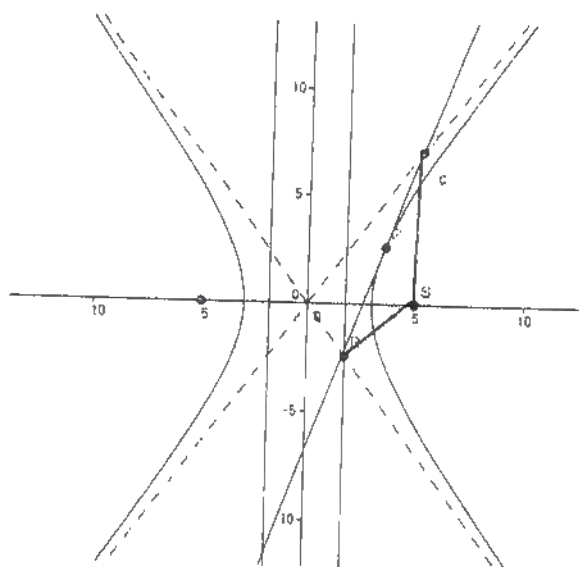
MATHEMATICS Extension 2: Question 9

Suggested Solutions

Marks

Marker's Comments

ii)



$$x = 3 \sec \theta \quad y = 4 \tan \theta$$

$$\frac{dx}{d\theta} = 3 \sec \theta \tan \theta \quad \frac{dy}{d\theta} = 4 \sec^2 \theta$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{4 \sec^2 \theta}{3 \sec \theta \tan \theta} \\ &= \frac{4 \sec \theta}{3 \tan \theta} \end{aligned}$$

$$y - 4 \tan \theta = \frac{4 \sec \theta}{3 \tan \theta} (x - 3 \sec \theta)$$

$$\begin{aligned} 3y \tan \theta - 12 \tan^2 \theta &= 4x \sec \theta - 12 \sec^2 \theta \\ 4x \sec \theta - 3y \tan \theta &= 12 \sec^2 \theta - 12 \tan^2 \theta \\ &= 12 (\sec^2 \theta - \tan^2 \theta) \\ &= 12 \quad \text{since } \sec^2 \theta - \tan^2 \theta = 1 \end{aligned}$$

$$\text{tangent } 4x \sec \theta - 3y \tan \theta = 12$$

Cannot quote formula need to derive it

Since show question need to make it clear what's happened to $\sec^2 \theta$ and $\tan^2 \theta$.

MATHEMATICS Extension 2: Question 9

Suggested Solutions

Marks

Marker's Comments

Tangent at P

$$4x \sec \theta - 3y \tan \theta = 12$$

$$\text{Let } y = \frac{4}{3}x$$

$$4x \sec \theta - 3 \times \frac{4}{3}x \tan \theta = 12$$

$$4x \sec \theta - 4x \tan \theta = 12$$

$$4x (\sec \theta - \tan \theta) = 12$$

$$x = \frac{3}{\sec \theta - \tan \theta}$$

$$y = \frac{4}{3} \times \frac{3}{\sec \theta - \tan \theta} = \frac{4}{\sec \theta - \tan \theta}$$

when for R at $y = -\frac{4}{3}x$

$$4x \sec \theta + 4x \tan \theta = 12$$

$$x = \frac{3}{\sec \theta + \tan \theta}$$

$$y = \frac{-4}{\sec \theta + \tan \theta}$$

mid point of RQ.

$$\left(\frac{\frac{3}{\sec \theta - \tan \theta} + \frac{3}{\sec \theta + \tan \theta}}{2}, \frac{\frac{4}{\sec \theta - \tan \theta} - \frac{4}{\sec \theta + \tan \theta}}{2} \right)$$

$$\left(\frac{3 \sec \theta + 3 \tan \theta + 3 \sec \theta - 3 \tan \theta}{2}, \frac{4 \sec \theta + 4 \tan \theta - 4 \sec \theta + 4 \tan \theta}{2} \right)$$

$$\left(\frac{6 \sec \theta}{2}, \frac{8 \tan \theta}{2} \right)$$

$(3 \sec \theta, 4 \tan \theta)$ which is point P

many students lost negative which made next part difficult

many students falsged answer

MATHEMATICS Extension 2: Question 9...

Suggested Solutions

Marks

Marker's Comments

$$iv) \text{ Area of triangle} = \frac{1}{2} \times d_{AK} \times P_{AK} + c_{AK}$$

$$\begin{aligned} \Delta_{AK} &= \sqrt{\left(\frac{3}{\sec\theta - \tan\theta} - \frac{3}{\sec\theta + \tan\theta}\right)^2 + \left(\frac{4}{\sec\theta + \tan\theta} + \frac{4}{\sec\theta - \tan\theta}\right)^2} \\ &= \sqrt{9 \left(\frac{\sec\theta + \tan\theta - \sec\theta + \tan\theta}{\sec^2\theta - \tan^2\theta}\right)^2 + 16 \left(\frac{\sec\theta + \tan\theta + \sec\theta - \tan\theta}{\sec^2\theta - \tan^2\theta}\right)^2} \\ &= \sqrt{36 \tan^2\theta + 64 \sec^2\theta} \\ &= 2 \sqrt{9 \tan^2\theta + 16 \sec^2\theta} \end{aligned}$$

1

$$\text{Perp dist} = \frac{|4x \sec\theta - 3y \tan\theta - 12|}{\sqrt{16 \sec^2\theta + 9 \tan^2\theta}}$$

$$= \frac{|-12|}{\sqrt{16 \sec^2\theta + 9 \tan^2\theta}}$$

$$\therefore \text{Area} = \frac{1}{2} \times \frac{12}{\sqrt{16 \sec^2\theta + 9 \tan^2\theta}} \times \sqrt{16 \sec^2\theta + 9 \tan^2\theta}$$

$$= \underline{12 u^2}$$

Other method
works
 $\frac{1}{2} \times 0 \times 0 \times \sin 60^\circ$
needed to
calculate
 $\tan^{-1} \frac{24}{7}$

MATHEMATICS Extension 2: Question...

Suggested Solutions

Marks

Marker's Comments

$$v) M_{QS} = \frac{4}{\sec\theta - \tan\theta} - 5$$

$$= \frac{4}{3 - 5(\sec\theta - \tan\theta)}$$

$$M_{RS} = \frac{-4}{3 - 5(\sec\theta + \tan\theta)}$$

$$\tan\theta = \left| \frac{\frac{4}{3 - 5(\sec\theta - \tan\theta)} + \frac{4}{3 - 5(\sec\theta + \tan\theta)}}{1 - \frac{16}{3 - 5(\sec\theta - \tan\theta)(3 - 5(\sec\theta + \tan\theta))}} \right|$$

$$= \left| \frac{12 - 20\sec\theta - 20\tan\theta + 12 - 20\sec\theta + 20\tan\theta}{9 - 15(\sec\theta + \tan\theta) - 15(\sec\theta - \tan\theta) + 25(\sec^2\theta - \tan^2\theta)} - 16 \right|$$

$$= \left| \frac{24 - 40\sec\theta}{9 + 25 - 16 - 30\sec\theta} \right|$$

$$= \left| \frac{4(6 - 10\sec\theta)}{3(6 - 10\sec\theta)} \right|$$

$$= \frac{4}{3}$$

Angle is obtuse

$$\therefore \angle QSR = 180 - 53^\circ 8'$$

$$= \underline{126^\circ 52'}$$

1 for both gradients